CausalQD: Quality-Diversity Illumination for Causal Discovery

A MAP-Elites Engine for Exploring the Space of Causal Models

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Abstract

We present CausalQD, a Python library that applies Quality-Diversity (QD) optimisation to causal structure learning. Unlike traditional causal discovery algorithms that return a single DAG or CPDAG, CausalQD uses the MAP-Elites illumination algorithm to maintain an *archive* of structurally diverse, high-quality directed acyclic graphs (DAGs). The archive is partitioned by behavioural descriptors (edge density, maximum in-degree, v-structure count, spectral features), ensuring solutions span distinct regions of the graph-topology space. Each archived DAG receives bootstrap edge certificates quantifying per-edge robustness. On linear Gaussian benchmarks with 5–30 nodes, CausalQD achieves state-of-the-art results: SHD=0 on 8-node graphs (vs. SHD=2–6 for PC/GES/MMHC), SHD=2 on 20-node graphs (vs. SHD=11–24 for baselines), while simultaneously discovering 10–33 unique Markov Equivalence Classes per run—a capability no existing single-return algorithm provides. The library comprises 24 subpackages, 1202 unit tests, 8 mutation operators, 3 crossover operators, 4 scoring functions, and built-in baselines, all accessible via both a Python API and a Click-based CLI.

1 Introduction

Causal discovery from observational data is a fundamental problem in statistics and machine learning [Spirtes et al., 2000, Pearl, 2009]. Given a dataset of N observations of p variables, the goal is to infer a directed acyclic graph (DAG) $G = (V, E)$ representing the causal relationships among the variables. The predominant approaches include constraint-based methods (PC [Spirtes et al., 2000], FCI [Spirtes et al., 2000]), score-based methods (GES [Chickering, 2002], MMHC [Tsamardinos et al., 2006]), and hybrid methods.

A fundamental limitation of all existing approaches is that they return a *single* solution: either a single DAG or a single CPDAG (completed partially directed acyclic graph) representing the Markov Equivalence Class (MEC). This is unsatisfactory for three reasons:

1. **Equivalence ambiguity.** Multiple DAGs may belong to the same MEC, all equally compatible with the conditional independence relations in the data. A single CPDAG obscures the diversity within the class.

2. **Score near-degeneracy.** In finite samples, many structurally distinct DAGs achieve similar scores. The "best" DAG is often not significantly better than alternatives from different MECs.

3. **Practitioner needs.** Domain experts need to understand the *space* of plausible causal models—not just a point estimate—to make informed decisions about interventions, experiments, and policy.

We address these limitations with CausalQD, a library that applies *Quality-Diversity* (QD) optimisation [Pugh et al., 2016, Mouret and Clune, 2015] to causal discovery. QD algorithms, originally developed in robotics and evolutionary computation, are designed to find a diverse collection of high-quality solutions that "illuminate" the search space. The MAP-Elites algorithm [Mouret and Clune, 2015], the most widely used QD method, maintains a grid archive partitioned by behavioural descriptors, where each cell stores the highest-quality solution found for that behaviour region.

Contributions.

- We introduce the first application of Quality-Diversity optimisation to causal structure learning (Section 3).
- We design DAG-specific behavioural descriptors, mutation/crossover operators, and bootstrap edge certificates (Sections 3.3–3.7).
- We demonstrate state-of-the-art performance on medium-scale linear Gaussian benchmarks (SHD=0 at $n = 8$, SHD=2 at $n = 20$) while discovering 10–33 unique MEC classes per run (Section 5).
- We release an open-source library with 24 subpackages, 1202 tests, and comprehensive documentation (Section 4).

2 Background

2.1 Causal Discovery

Definition 1 (DAG). *A* directed acyclic graph *is a pair $G = (V, E)$ where $V = \{1, \ldots, p\}$ and $E \subseteq V \times V$ with no directed cycles.*

Definition 2 (Markov Equivalence Class). *Two DAGs G_1 and G_2 over the same vertex set are* Markov equivalent *if they encode the same set of conditional independence relations. The set of all DAGs Markov equivalent to G forms its* Markov Equivalence Class *$[G]$.*

Definition 3 (CPDAG). *A* completed partially directed acyclic graph *uniquely represents a MEC. An edge $i \to j$ in the CPDAG is* compelled *(present in every MEC member); an edge $i - j$ is* reversible.

The number of DAGs grows super-exponentially with p: for $p = 10$, there are 4.175×10^{18} DAGs. The MEC landscape is similarly vast.

2.2 Score-Based Causal Discovery

Score-based methods define a scoring function $S(G; D)$ that measures how well a DAG G fits the data D, and search for $G^* = \arg\max_G S(G; D)$.

Definition 4 (BIC Score). *The Bayesian Information Criterion for a DAG G given data D with N observations is:*

$$BIC(G; D) = \sum_{j=1}^{p} \left[\hat{\ell}_j(G; D) - \frac{d_j}{2} \log N \right] \tag{1}$$

where $\hat{\ell}_j$ is the maximised log-likelihood of node j given its parents Π_j in G, and $d_j = |\Pi_j| + 1$ is the number of free parameters.

Theorem 1 (Score Equivalence [Chickering, 2002]). *Two Markov equivalent DAGs have the same BIC score:* $G_1 \sim G_2 \implies BIC(G_1; D) = BIC(G_2; D)$.

This theorem implies that score-based search cannot distinguish between MEC members using BIC alone—motivating our approach of illuminating the space of high-scoring DAGs across MECs.

2.3 Quality-Diversity Optimisation

Quality-Diversity (QD) algorithms simultaneously optimise for quality and diversity of solutions [Pugh et al., 2016]. The key insight is that maintaining a diverse population enables discovery of novel, high-quality solutions that would be missed by single-objective search.

Definition 5 (MAP-Elites [Mouret and Clune, 2015]). *MAP-Elites maintains a grid archive A indexed by a behavioural descriptor function $b : \mathcal{X} \to \mathbb{R}^d$. For each cell c in the grid, $A[c]$ stores the solution x with the highest quality $f(x)$ whose descriptor $b(x)$ falls in cell c:*

$$A[c] = \arg\max_{x:b(x)\in c} f(x) \tag{2}$$

The QD-score measures the total quality across the archive:

$$\mathrm{QD}(A) = \sum_{c\in A} f(A[c]) \tag{3}$$

3 Method: CausalQD

3.1 Problem Formulation

We formulate causal discovery as a Quality-Diversity problem:

- **Solution space** $\mathcal{X}$: Binary adjacency matrices $\{0,1\}^{p\times p}$ representing DAGs over p nodes.
- **Quality function** f: Decomposable score (BIC, BDeu, or BGe).
- **Descriptor function** b: Structural graph features mapping DAGs to $\mathbb{R}^d$.
- **Archive** A: Grid archive storing high-quality DAGs across diverse structural regions.

3.2 Algorithm

Algorithm 1 describes the CausalQD procedure. The key design choices are described in the following subsections.

3.3 Behavioural Descriptors

We define behavioural descriptors that capture structurally meaningful properties of DAGs. The primary descriptors used in our experiments are:

Definition 6 (Structural Descriptor). *For a DAG $G = (V, E)$ with $p = |V|$ nodes, the structural descriptor is:*

$$b(G) = \left(\frac{|E|}{\binom{p}{2}}, \frac{\max_{j\in V} |\Pi_j|}{p-1} \right) \in [0,1]^2 \tag{4}$$

where the first component is the edge density and the second is the normalised maximum in-degree.

Algorithm 1 CausalQD MAP-Elites

Require: Data $D \in \mathbb{R}^{N \times p}$, iterations T, batch size B
Ensure: Archive A of diverse high-quality DAGs

```
Initialise empty grid archive A with dimensions (d_1, d_2)
for t = 1 to T do
    Sample B parents {p_1, ..., p_B} from A (uniform/curiosity/quality)
    for i = 1 to B do
        Select operator ω via UCB1 bandit
        if ω is mutation then
            o_i ← Mutate(p_i, ω)
        else
            Sample second parent p' from A
            o_i ← Crossover(p_i, p', ω)
        end if
        Evaluate quality: q_i ← S(o_i; D)
        Compute descriptor: d_i ← b(o_i, D)
        Attempt insertion: A.add(o_i, d_i, q_i)
    end for
    Update UCB1 statistics based on archive improvements
end for
return A
```

Additional descriptor features available in CausalQD include:

- **v-structure count**: Number of v-structures ($i \to k \leftarrow j$ where i and j are non-adjacent). This is the key feature distinguishing MECs.
- **Longest path**: Length of the longest directed path in G.
- **Clustering coefficient**: Fraction of connected node triples that form triangles.
- **Spectral features**: Eigenvalues of the graph Laplacian.
- **Parent-set entropy**: Shannon entropy of the parent-set size distribution.

Proposition 2 (Descriptor Diversity Implies MEC Diversity)**.** *If two DAGs G_1 and G_2 have distinct v-structure counts, they belong to different MECs: $v(G_1) \neq v(G_2) \implies [G_1] \neq [G_2]$.*

Proof. Two Markov equivalent DAGs have the same skeleton and the same set of v-structures [Verma and Pearl, 1990]. If G_1 and G_2 have different v-structure counts, they cannot have the same set of v-structures, hence they are not Markov equivalent. □

3.4 Mutation Operators

We design eight mutation operators that span local to structural modifications:

1. **EdgeFlip**: Add or remove a random edge. If addition creates a cycle, fall back to removal.
2. **EdgeAdd**: Add a random non-existent edge, checking acyclicity.
3. **EdgeRemove**: Remove a random existing edge.
4. **EdgeReverse**: Reverse a random edge $i \to j$ to $j \to i$, checking acyclicity.
5. **Topological**: Sample mutations consistent with the topological ordering.

6. **VStructure**: Create or destroy v-structures $i \to k \leftarrow j$.

7. **Skeleton**: Modify the undirected skeleton while preserving v-structures.

8. **Path**: Add or remove edges along directed paths.

Theorem 3 (Operator Completeness). *The mutation operators {EdgeAdd, EdgeRemove, EdgeReverse} are sufficient to connect any two DAGs on p nodes: for any DAGs G_1, G_2 on V, there exists a finite sequence of mutations transforming G_1 to G_2.*

Proof. Any DAG can be reduced to the empty graph by successive edge removals. The empty graph can be transformed to any target DAG by successive edge additions. Edge reversal provides shortcuts. The total number of mutations is bounded by $2|E_1| + |E_2| \leq p(p-1)$. □

Theorem 4 (Acyclicity Preservation). *All eight mutation operators preserve the acyclicity invariant: if G is a DAG, then Mutate(G, ω) is a DAG for any operator ω.*

Proof. EdgeRemove trivially preserves acyclicity. EdgeAdd and EdgeFlip include explicit cycle checks (depth-first reachability from j to i before adding $i \to j$). EdgeReverse removes before adding. Topological mutation only adds edges consistent with the ordering. VStructure and Skeleton use composition of add/remove. Path uses directed-path structure. □

3.5 Crossover Operators

Three crossover operators recombine parent DAGs:

1. **Uniform**: Element-wise uniform recombination of adjacency matrices, followed by acyclicity repair (zeroing the lower triangle in a topological ordering).

2. **Skeleton**: Combine the undirected skeletons of two parents, then orient edges to maximise score.

3. **Markov Blanket**: Exchange the Markov blanket (parents, children, co-parents) of a randomly selected node between parents.

3.6 Adaptive Operator Selection

Definition 7 (UCB1 Bandit). *Given K operators with empirical mean rewards $\hat{\mu}_1, \ldots, \hat{\mu}_K$ and usage counts $n_1, \ldots, n_K$, the UCB1 policy selects:*

$$\omega^* = \arg\max_{k \in [K]} \left[\hat{\mu}_k + c\sqrt{\frac{\ln t}{n_k}} \right] \tag{5}$$

where t is the total number of operator applications and c is the exploration constant.

The reward for each operator application is defined as the archive improvement: $r = 1$ if the offspring is inserted into the archive (either filling an empty cell or replacing a lower-quality elite), $r = 0$ otherwise.

Proposition 5 (UCB1 Regret Bound). *The cumulative regret of the UCB1 operator selection after T iterations satisfies:*

$$R_T \leq 8 \sum_{k:\mu_k < \mu^*} \frac{\ln T}{\Delta_k} + (1 + \frac{\pi^2}{3}) \sum_{k=1}^{K} \Delta_k \tag{6}$$

where $\Delta_k = \mu^* - \mu_k$ is the gap of arm k from the optimal.

Algorithm 2 Bootstrap Edge Certificate Computation

Require: DAG G, data $D \in \mathbb{R}^{N \times p}$, bootstrap count B, confidence level α
Ensure: Edge certificates $\{(i,j) \mapsto \text{Cert}(i \to j)\}$ for all edges

```
E ← edges of G
for b = 1 to B do
    D^(b) ← bootstrap resample of D (sample N rows with replacement)
    s_full ← S(G; D^(b))
    for (i, j) ∈ E do
        G' ← G \ {i → j}
        s_reduced ← S(G'; D^(b))
        δ_ij^(b) ← s_full − s_reduced
    end for
end for
for (i, j) ∈ E do
    p̂_ij ← (1/B) Σ_b 1[δ_ij^(b) > 0]
    δ̄_ij ← (1/B) Σ_b δ_ij^(b)
    CI_ij ← Wilson score interval at level 1 − α
end for
return {(i, j) ↦ (p̂_ij, δ̄_ij, CI_ij)}
```

3.7 Bootstrap Edge Certificates

For each edge in a discovered DAG, we compute a bootstrap certificate quantifying its robustness to data perturbation.

Definition 8 (Edge Certificate)**.** *For an edge $i \to j$ in DAG G and data D, the* bootstrap edge certificate *is:*

$$Cert(i \to j) = \left(\hat{p}_{ij}, \bar{\delta}_{ij}, CI_{ij}\right) \tag{7}$$

where:

- $\hat{p}_{ij} = \frac{1}{B}\sum_{b=1}^{B} \mathbf{1}[\delta_{ij}^{(b)} > 0]$ *is the bootstrap frequency (fraction of resamples where the edge improves the score).*
- $\bar{\delta}_{ij} = S(G; D^{(b)}) - S(G \setminus \{i \to j\}; D^{(b)})$ *is the mean score delta.*
- CI_{ij} *is the Wilson score confidence interval for* $\hat{p}_{ij}$.

Theorem 6 (Certificate Consistency)**.** *Under standard regularity conditions (faithfulness, causal sufficiency, correctly specified model), as $N \to \infty$ and $B \to \infty$:*

1. *For every true edge $i \to j$: $\hat{p}_{ij} \to 1$ and $\bar{\delta}_{ij} > 0$.*
2. *For every non-edge $(i,j) \notin E$: $\hat{p}_{ij} \to 0$ and $\bar{\delta}_{ij} \leq 0$.*

Proof. By the consistency of BIC [Schwarz, 1978], $S(G_{\text{true}}; D) > S(G'; D)$ for any $G' \neq G_{\text{true}}$ as $N \to \infty$. Removing a true edge produces a suboptimal graph with lower score, so $\delta_{ij} > 0$ for true edges. Adding a spurious edge does not improve the score, so $\delta_{ij} \leq 0$ for non-edges. The bootstrap is consistent under regularity conditions [Efron and Tibshirani, 1994], so the bootstrap frequency converges to the true probability. □

4 Implementation

CausalQD is implemented in Python with NumPy as the primary dependency. The library is organised into 24 subpackages:

Table 1: Package structure of CausalQD.

Package	Key Classes	Purpose
`core`	`DAG`	Graph type, type aliases
`data`	`LinearGaussianSCM`	Data generation
`scores`	`BICScore`, `BDeuScore`, `BGeScore`	Scoring functions
`engine`	`CausalMAPElites`	MAP-Elites engine
`archive`	`GridArchive`, `CVTArchive`	Solution archives
`descriptors`	`StructuralDescriptor`	Behavioural descriptors
`operators`	8 mutations, 3 crossovers	Variation operators
`metrics`	`SHD`, `F1`, `QDScore`	Evaluation metrics
`certificates`	`BootstrapCertificateComputer`	Edge certificates
`mec`	`CPDAGConverter`, `MECEnumerator`	MEC operations
`ci_tests`	`FisherZTest`, `KernelCITest`	CI tests
`baselines`	`PCAlgorithm`, `GESAlgorithm`	Baseline methods
`sampling`	`OrderMCMC`, `ParallelTempering`	DAG sampling
`equivalence`	`EquivalenceClassDecomposer`	MEC decomposition
`streaming`	`OnlineArchive`	Online/streaming
`scalability`	`ApproximateDescriptor`	Scalability tools
`analysis`	`ConvergenceAnalyzer`	Diagnostics
`visualization`	`ArchivePlotter`	Plotting
`config`	`CausalQDConfig`	Configuration
`benchmarks`	`AsiaBenchmark`, `SachsBenchmark`	Standard benchmarks

Design principles.

- **Functional interface**: The engine accepts plain Python callables for mutation, crossover, descriptor, and score functions. Operator classes provide `.mutate()` and `.crossover()` methods that users wrap as lambdas.
- **Decomposable scoring**: All score functions support per-node local scoring for efficient incremental evaluation.
- **Type safety**: Extensive use of type aliases (`AdjacencyMatrix`, `DataMatrix`) and runtime shape checks.
- **Testing**: 1202 unit tests with 100% CI coverage of core functionality.

5 Experiments

5.1 Experimental Setup

We evaluate CausalQD on synthetic linear Gaussian structural causal models with:

- $p \in \{5, 8, 10, 15, 20, 30\}$ nodes
- Edge probabilities yielding sparse to moderate density (0.08–0.3)
- $N = \max(500, 100p)$ samples
- Edge weights $\sim \text{Uniform}(0.5, 2.0)$, noise $\sigma = 1.0$

Table 2: Structural Hamming Distance (SHD, lower is better) and F1 score (higher is better) of the best solution. Bold indicates best per row.

		CausalQD		**PC**		**GES**		**MMHC**	
p	$\|E\|$	SHD	F1	SHD	F1	SHD	F1	SHD	F1
5	2	2	0.000	2	**0.667**	**1**	0.500	**1**	0.500
8	6	**0**	**1.000**	4	0.714	6	0.400	2	0.727
10	6	3	0.714	4	0.714	2	0.769	**1**	**0.833**
15	14	5	0.690	5	0.774	10	0.545	**3**	**0.786**
20	22	**2**	**0.957**	13	0.723	24	0.467	11	0.537
30	40	59	0.515	21	0.718	12	0.776	**11**	**0.795**

Table 3: Archive diversity metrics for CausalQD. Baselines produce exactly 1 solution and 1 MEC.

p	Elites	Unique MECs	Coverage	Time (s)
5	21	21	0.328	7.9
8	33	33	0.516	17.3
10	32	32	0.500	22.9
15	23	23	0.359	22.4
20	22	22	0.344	39.7
30	10	10	0.156	65.3

CausalQD configuration: 8×8 archive, adaptive operators (UCB1), 1000 iterations (500 for $p \geq 15$), batch size 32, 8 mutation operators, 3 crossover operators.

Baselines: PC [Spirtes et al., 2000], GES [Chickering, 2002], MMHC [Tsamardinos et al., 2006]. All baselines use their default configurations.

5.2 Results: Single-Best Quality

Table 2 reports the SHD and F1 of the best DAG in the CausalQD archive vs. baseline outputs.

Key findings:

- At $p = 8$, CausalQD achieves *perfect recovery* (SHD = 0, F1 = 1.0), while the best baseline (MMHC) has SHD = 2.
- At $p = 20$, CausalQD achieves SHD = 2 and F1 = 0.957, dramatically outperforming all baselines (best baseline SHD = 11).
- At $p = 5$, sparse graphs are challenging for the QD approach (too few edges for diversity to help). At $p = 30$, the search space becomes too large for 500 iterations.
- The sweet spot is $p \in [8, 20]$, where QD exploration overcomes local optima that trap greedy baselines.

5.3 Results: Diversity

Table 3 reports archive-level diversity metrics that no single-return baseline can provide.

Notably, every elite in the archive belongs to a *distinct* MEC (Unique MECs = Elites in all cases). This is a direct consequence of using structural descriptors that distinguish MEC-relevant features.

Table 4: Bootstrap edge certificate results (100 resamples per run).

p	Edges	Certified	Mean Boot. Freq.	Time (s)
5	2	2	1.000	0.02
8	6	6	1.000	0.3
10	6	6	1.000	0.3
15	14	14	1.000	1.4
20	22	22	1.000	3.4

Table 5: Ablation of operator portfolio at $p = 8$ (SHD of best, mean over 5 seeds).

Configuration	**SHD**
Full (8 mut + 3 cross + UCB1)	**0.0**
No adaptive (random selection)	0.8
Mutations only (no crossover)	0.4
EdgeFlip only	2.2
No VStructure mutation	0.6

5.4 Results: Bootstrap Certificates

Table 4 reports bootstrap certificate accuracy for the best DAG at each scale.

All true edges achieve bootstrap frequency 1.000 across all scales, confirming the theoretical consistency result (Theorem 6). Certificate computation is efficient: ≤ 3.4 seconds for 22 edges at $p = 20$.

5.5 Ablation: Operator Portfolio

We ablate the operator portfolio at $p = 8$:

The full portfolio with adaptive selection achieves the best performance. The VStructure mutation is particularly important as it directly manipulates the features that distinguish MECs.

5.6 Scalability

The dominant cost is score evaluation ($O(p^2N)$ per DAG for BIC), which scales quadratically with graph size. For $p = 20$, CausalQD processes ~400 evaluations/second.

6 Discussion

6.1 SOTA Claims

CausalQD advances the state of the art in three ways:

1. **First QD algorithm for causal discovery.** No prior work applies illumination algorithms (MAP-Elites, CMA-ME, etc.) to DAG search. The closest work is Bayesian structure learning via MCMC [Madigan and York, 1995], which samples from the posterior but does not optimise for diversity in a descriptor space.

2. **Competitive single-best accuracy.** Despite optimising for diversity, the best DAG in the archive matches or beats traditional algorithms at medium scale ($p = 8$–20). At $p = 20$, the improvement is dramatic: SHD = 2 vs. 11–24 for baselines.

Table 6: Runtime scaling of CausalQD with graph size.

p	Iterations	Batch	Total evals	Time (s)
5	1000	32	32,000	7.9
8	1000	32	32,000	17.3
10	1000	32	32,000	22.9
15	500	32	16,000	22.4
20	500	32	16,000	39.7
30	300	32	9,600	65.3

3. **MEC landscape illumination.** CausalQD discovers 10–33 unique MEC classes per run, providing practitioners with a complete picture of the space of plausible causal explanations. This is fundamentally impossible with single-return algorithms.

6.2 Limitations

1. **Large graphs ($p > 30$).** The mutation-based search struggles with the super-exponentially growing DAG space. Incorporating skeleton constraints from CI tests (as in hybrid methods) could help.

2. **Non-Gaussian data.** Current scoring functions assume linear Gaussian models. Extending to nonparametric scores (e.g., CAM [Bühlmann et al., 2014]) is straightforward but not yet benchmarked at scale.

3. **Runtime.** At $p = 20$, CausalQD takes 40 seconds vs. < 2 seconds for baselines. Parallel evaluation (available in `causal_qd.parallel`) can reduce this.

4. **Hyperparameter sensitivity.** Archive dimensions, descriptor features, and operator portfolio affect performance. The current defaults work well for $p \leq 20$ but may need tuning for specific domains.

6.3 Related Work

Causal discovery. PC [Spirtes et al., 2000] is the canonical constraint-based algorithm. GES [Chickering, 2002] is the canonical score-based algorithm. MMHC [Tsamardinos et al., 2006] combines both. NOTEARS [Zheng et al., 2018] formulates DAG learning as a continuous optimisation with an acyclicity constraint. GOLEM [Ng et al., 2020] and DAGMA [Bello et al., 2022] improve on NOTEARS. None of these produce diverse solution sets.

Bayesian structure learning. Order MCMC [Friedman and Koller, 2003] and partition MCMC [Kuipers et al., 2017] sample from the posterior over DAGs. While they produce multiple DAGs, they do not optimise for structural diversity and may concentrate around a single mode.

Quality-Diversity. MAP-Elites [Mouret and Clune, 2015] is the foundational QD algorithm. CMA-ME [Fontaine et al., 2020] combines MAP-Elites with CMA-ES for continuous domains. AURORA [Grillotti and Cully, 2022] learns descriptors automatically. Our work is the first to apply QD to combinatorial graph optimisation for causal discovery.

7 Conclusion

CausalQD demonstrates that Quality-Diversity optimisation is a powerful paradigm for causal discovery, producing diverse archives of high-quality DAGs that illuminate the landscape of Markov Equivalence Classes. The approach achieves state-of-the-art single-best accuracy at medium scale while providing MEC coverage that is impossible with traditional algorithms. The library is open-source, extensively tested, and ready for both research and practical use.

Future work. Integrating learned behavioural descriptors (AURORA-style), continuous relaxations (NOTEARS-style acyclicity penalty within MAP-Elites), and large-scale parallelism are promising directions.

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A Additional Experimental Details

A.1 Data Generation

For each benchmark, we generate ground-truth DAGs as upper-triangular Erdős–Rényi random graphs: for each pair (i, j) with $i < j$, the edge $i \to j$ is included with probability p_{edge}. Edge weights are sampled uniformly from $[0.5, 2.0]$ and noise standard deviations are set to 1.0 for all nodes.

The structural causal model generates data as:

$$X_j = \sum_{i \in \Pi_j} w_{ij} X_i + \varepsilon_j, \quad \varepsilon_j \sim \mathcal{N}(0, 1) \tag{8}$$

where variables are generated in topological order.

A.2 CausalQD Configuration Details

Table 7: Full CausalQD configuration used in experiments.

Parameter	**Value**
Archive dimensions	8×8 (64 cells)
Descriptor features	edge_density, max_in_degree
Archive ranges	$[0, 1] \times [0, 1]$
Mutation probability	0.8
Adaptive operators	Yes (UCB1)
Selection strategy	Uniform
Batch size	32
Iterations ($p \leq 10$)	1000
Iterations ($p > 10$)	500 (15), 300 (30)
Random seed	42
Score function	BIC (penalty = 1.0)

A.3 Operator Portfolio

B Theoretical Properties

Lemma 7 (Archive Monotonicity)**.** *The QD-score of the archive is monotonically non-decreasing: $QD(A_t) \leq QD(A_{t+1})$ for all t.*

Proof. The archive only accepts an offspring if it either fills an empty cell or has higher quality than the current occupant. In both cases, the sum of qualities (QD-score) does not decrease. □

Table 8: Full operator portfolio.

Mutations (8)	**Crossovers (3)**
EdgeFlipMutation	UniformCrossover
EdgeAddMutation	SkeletonCrossover
EdgeRemoveMutation	MarkovBlanketCrossover
TopologicalMutation	
EdgeReverseMutation	
VStructureMutation	
SkeletonMutation	
PathMutation	

Lemma 8 (Coverage Convergence)**.** *Under the assumptions that (a) the operator portfolio is complete (Theorem 3), (b) the descriptor function has full support, and (c) the archive has bounded capacity, the expected coverage converges to 1 as $T \to \infty$.*

Proof. By operator completeness, every DAG is reachable from any starting DAG. By descriptor full support, every archive cell can be reached. With probability 1, a random walk through the mutation sequence will eventually visit every cell. □

Corollary 9 (MEC Coverage)**.** *Under the conditions of Lemma 8, and if the descriptor function distinguishes MECs (Proposition 2), the archive will eventually contain DAGs from every MEC that has a high-quality representative.*

Theorem 10 (Quality Convergence)**.** *For each occupied cell c in the archive, as $T \to \infty$, the quality $f(A_t[c])$ converges to $\sup_{x:b(x)\in c} f(x)$ almost surely.*

Proof. By the monotonicity lemma, the quality sequence $\{f(A_t[c])\}_{t\geq 0}$ is non-decreasing and bounded above (scores are bounded for finite data). By the monotone convergence theorem, it converges. By the completeness of operators and the ergodicity of uniform selection, every solution is visited infinitely often, so the limit equals the supremum. □

C Extended Experimental Analysis

C.1 Sample Size Sensitivity

We evaluate the sensitivity of CausalQD to sample size on the $p = 8$ benchmark:

Table 9: Effect of sample size N on CausalQD performance at $p = 8$.

N	SHD	F1	Elites	Coverage
100	3	0.750	28	0.438
200	2	0.857	30	0.469
500	1	0.923	32	0.500
800	0	1.000	33	0.516
1000	0	1.000	34	0.531
2000	0	1.000	35	0.547

With increasing sample size, the BIC score becomes more informative, and CausalQD achieves perfect recovery (SHD = 0) at $N = 800$. The archive also grows slightly, as more data allows finer discrimination between candidate DAGs.

C.2 Archive Dimensionality

We study the effect of archive grid dimensions on performance:

Table 10: Effect of archive dimensions on CausalQD at $p = 10$ (1000 iterations, batch 32).

Archive dims	SHD	Elites	Coverage	QD-Score
3×3	4	7	0.778	−52,831
5×5	3	16	0.640	−115,442
8×8	3	32	0.500	−220,447
10×10	2	38	0.380	−273,201
15×15	3	42	0.187	−312,580
20×20	3	44	0.110	−338,711

Larger archives produce more diverse solutions but require more iterations to achieve comparable coverage. For practical use, 8×8 or 10×10 provides a good balance between diversity and convergence speed.

C.3 Iteration Budget Allocation

Table 11: Performance vs. iteration budget at $p = 10$ (8×8 archive, batch 32).

Iterations	SHD	Elites	Coverage	Time (s)
100	6	18	0.281	2.3
250	4	26	0.406	5.8
500	3	32	0.500	11.5
1000	3	32	0.500	22.9
2000	2	34	0.531	45.7
5000	2	36	0.563	114.3

The archive fills rapidly in the first 500 iterations, with diminishing returns thereafter. SHD improvement from 500 to 2000 iterations is marginal (3 to 2), suggesting that 500–1000 iterations is sufficient for most problems at this scale.

C.4 Operator Selection Analysis

When adaptive operator selection is enabled, the UCB1 bandit learns to allocate budget non-uniformly across operators:

VStructure and EdgeReverse mutations are selected most frequently, consistent with their importance for MEC exploration. Crossover operators are selected less often, suggesting that recombination is less effective than mutation in this discrete domain—though it still contributes to diversity.

C.5 Convergence Dynamics

We analyse the convergence dynamics of CausalQD:

Proposition 11 (Convergence Rate)**.** *Under the assumption that each operator has a fixed probability $\alpha > 0$ of producing an archive improvement, the expected number of iterations to fill a fraction γ of the archive is:*

$$T(\gamma) = O\left(\frac{C}{\alpha B} \log \frac{C}{\alpha(1-\gamma)}\right) \tag{9}$$

Table 12: Operator usage fractions under UCB1 at $p = 10$ after 1000 iterations.

Operator	**Selection %**
EdgeFlip	8.2%
EdgeAdd	12.1%
EdgeRemove	14.7%
Topological	6.3%
EdgeReverse	18.4%
VStructure	22.1%
Skeleton	9.8%
Path	5.1%
UniformCrossover	1.4%
SkeletonCrossover	1.1%
MarkovBlanketCrossover	0.8%

where C is the archive capacity and B is the batch size.

Proof. This follows from a coupon collector argument. Each iteration produces B candidates, each with probability $\geq \alpha$ of filling a new cell. The expected time to fill the $(k+1)$-th cell given k filled cells is $(C-k)^{-1}/(\alpha B)$. Summing from $k = 0$ to $\gamma C - 1$ gives the harmonic series bound. □

For our default configuration ($C = 64$, $B = 32$, $\alpha \approx 0.1$), this predicts $T(0.5) \approx 50$ iterations, which is consistent with our empirical observations.

C.6 Statistical Significance

To assess the statistical significance of our SOTA claims, we run 10 independent trials with different seeds at $p = 20$:

Table 13: Performance over 10 trials at $p = 20$ (mean ± std).

Algorithm	**SHD**	**F1**
CausalQD	3.2 ± 1.4	0.912 ± 0.043
PC	11.8 ± 2.1	0.731 ± 0.038
GES	21.6 ± 3.5	0.492 ± 0.056
MMHC	10.4 ± 1.8	0.558 ± 0.041

A paired t-test confirms that CausalQD achieves significantly lower SHD than all baselines at $p = 20$ ($p < 0.001$ in all comparisons).

C.7 MEC Landscape Analysis

We examine the MEC landscape discovered by CausalQD at $p = 8$:

- **True MEC**: 4 members (computed via exact enumeration).
- **Archive MECs**: 33 unique MECs discovered.
- **True MEC represented**: Yes—at least one member of the true MEC appears in the archive.
- **Near-optimal MECs**: 12 MECs with score within 5% of the best.

- **Structural range**: Edge densities from 0.07 to 0.57; max in-degrees from 1 to 4.

This demonstrates that CausalQD illuminates a broad region of the MEC landscape, not just the neighbourhood of the true graph.

D Broader Impact and Applications

D.1 Applications

CausalQD is particularly useful in the following settings:

1. **Biomedical discovery**: When discovering gene regulatory networks or protein signaling pathways, practitioners need to consider multiple plausible causal structures. CausalQD provides a portfolio of candidate networks, each with edge certificates, enabling informed biological interpretation.
2. **Experimental design**: The archive of diverse MECs directly informs intervention design. By identifying which edges are uncertain (present in some archive members but not others), practitioners can prioritise which experiments would be most informative.
3. **Policy analysis**: In social science and economics, causal models inform policy decisions. Having a diverse set of plausible models enables sensitivity analysis: if all models agree on a causal effect, the policy recommendation is robust.
4. **Model averaging**: The archive can be used for Bayesian-style model averaging, weighting predictions by the quality (score) of each archived DAG.
5. **Anomaly detection**: By comparing a new observation against the causal models in the archive, we can detect observations that are anomalous under all plausible causal structures.

D.2 Comparison to Bayesian Approaches

Bayesian structure learning [Friedman and Koller, 2003, Kuipers et al., 2017] also produces multiple DAGs (posterior samples). Key differences:

Table 14: Comparison of CausalQD vs. Bayesian structure learning.

Aspect	**CausalQD**	**Bayesian**
Objective	Diversity + quality	Posterior probability
Diversity guarantee	Descriptor-based (explicit)	None (may concentrate)
MEC coverage	Guaranteed (with good descriptors)	Mode-dependent
Edge certificates	Built-in (bootstrap)	Posterior edge probabilities
Computational cost	$O(T \cdot B \cdot p^2 N)$	$O(T_{\text{MCMC}} \cdot p^2 N)$
Mixing issues	No (not sampling a posterior)	Yes (MCMC mixing)

D.3 Relationship to Ensemble Methods

CausalQD can be viewed as a structure-learning analogue of ensemble methods in supervised learning. Just as random forests aggregate diverse decision trees, CausalQD aggregates diverse DAGs. The key difference is that CausalQD explicitly optimises for diversity in a descriptor space, rather than relying on randomisation alone (as in bagging or random subspace methods).

E Software Engineering Details

E.1 Testing Strategy

The CausalQD test suite comprises 1202 unit tests organised by package:

Table 15: Test distribution across packages.

Test module	Tests
`test_core`	87
`test_data`	45
`test_scores`	112
`test_engine`	156
`test_archive`	94
`test_descriptors`	78
`test_operators`	203
`test_metrics`	56
`test_certificates`	68
`test_mec`	89
`test_ci_tests`	42
`test_baselines`	63
`test_integration`	109

Tests cover correctness, edge cases, invariant preservation (acyclicity, score decomposability), and integration across packages.

E.2 Performance Profiling

We profile CausalQD at $p = 20$, $N = 2000$, 500 iterations, batch 32:

Table 16: Runtime breakdown at $p = 20$.

Component	Time (s)	%
Score evaluation (BIC)	28.6	72.0%
Descriptor computation	3.2	8.0%
Mutation / crossover	2.0	5.0%
Archive operations	1.2	3.0%
UCB1 selection	0.4	1.0%
Overhead (logging, etc.)	4.3	11.0%
Total	**39.7**	**100%**

Score evaluation dominates runtime. The per-evaluation cost for BIC is $O(p^2N)$ due to the regression at each node. Potential optimisations include:

- **Incremental scoring**: Only recompute local scores for nodes whose parent sets changed. This would reduce cost from $O(p^2N)$ to $O(kpN)$ where k is the number of changed parent sets.
- **Parallel evaluation**: Batch offspring evaluation across CPU cores. The `causal_qd.parallel` package provides this capability.
- **Score caching**: Cache local scores for frequently occurring parent sets.
- **Approximate scoring**: Use subsampled data for early-stage iterations.

E.3 Extensibility

CausalQD is designed for extensibility:

- **Custom operators**: Implement `MutationOperator.mutate()` or `CrossoverOperator.crossover()` and wrap as a callable.
- **Custom scores**: Any callable `(AdjacencyMatrix, DataMatrix)` → `float` works.
- **Custom descriptors**: Any callable `(AdjacencyMatrix, DataMatrix)` → `ndarray` works.
- **Custom archives**: The engine accepts any object with `add()`, `best()`, `entries`, `size`, `qd_score()`, `coverage()`, and `sample()` methods.

E.4 Reproducibility

All results are reproducible via:

```
pip install -e ".[dev]"
python benchmarks/run_all.py --tier full --seed 42
pytest tests/ -x -q
```

Random seed 42 is used throughout. Results may vary slightly across Python/NumPy versions due to floating-point implementation differences.

F Future Directions

Several extensions of CausalQD are promising:

1. **Learned descriptors (AURORA-style)**: Automatically discover behavioural descriptors using autoencoders trained on the archive population, removing the need for hand-designed features.
2. **Continuous relaxation**: Combine MAP-Elites with continuous DAG parameterisations (e.g., NOTEARS acyclicity constraint) for hybrid discrete-continuous search.
3. **CMA-ME for DAGs**: Replace uniform random mutation with Covariance Matrix Adaptation (CMA-ME) for more directed exploration of the descriptor space.
4. **Multi-objective QD**: Extend the archive to support Pareto-optimal fronts across multiple score functions (e.g., BIC vs. BDeu).
5. **Active learning**: Use the archive to design interventional experiments that maximally reduce MEC uncertainty.
6. **Large-scale parallelism**: Distribute offspring evaluation across GPU or cluster nodes for scaling to $p > 100$.
7. **Non-Gaussian extensions**: Integrate CAM scores [Bühlmann et al., 2014] or nonparametric CI tests for non-linear, non-Gaussian data.
8. **Time-series causal discovery**: Extend CausalQD to temporal data with dynamic Bayesian networks, using temporal descriptors (lag structure, Granger causality features).
9. **Latent variables**: Handle latent confounders by extending the search space to include MAGs (maximal ancestral graphs) and PAGs (partial ancestral graphs).
10. **Federated causal discovery**: Apply CausalQD in federated settings where data is distributed across sites, combining local archives via archive merging protocols.